

Day 11 of training

JavaScript

Events in JavaScript

JavaScript interaction with HTML is handled through events that occur when the user or the browser manipulates a page. When the page loads, it is called an event. When the user clicks a button, that click too is an event.

->click -> Key press
->Double click -> Key up
-->right click ->load ----|
->Mouse Hover ->unload |. these are window events
->MOuse out ->resize |
->Mouse up ->scroll -----|

Mouse Events in JavaScript

Mouse events are actions that happen when the user interacts with a mouse on a web page — like clicking, hovering, moving, etc.

Event Name	What It Means (Easy)
-----	-----
`click`	When you click (press and release) the mouse
`dblclick`	When you double-click the mouse
`mousedown`	When you press down the mouse button
`mouseup`	When you release the mouse button
`mousemove`	When you move the mouse over an element
`mouseenter`	When the mouse enters (comes over) an element
`mouseleave`	When the mouse leaves the element
`mouseover`	Like `mouseenter` but also triggers for child elements
`mouseout`	Like `mouseleave` but also triggers for child elements

Key Events in JavaScript

Key events happen when you press or release a key on your keyboard.

JavaScript can detect and respond to those actions — like when you type something.

Event Name	What It Means (Easy)
-----	-----
`keydown`	When you press down a key
`keypress`	When you press a key that gives a character (older, now rarely used)
`keyup`	When you release the key

Window + Key Events in JavaScript

You can attach keyboard events to the whole window object, so it listens to key presses anywhere on the page, not just in a specific input box.

Event	What It Means (Easy)
-----	-----
`load`	Page fully loaded (images, styles too)
`resize`	When the browser window size changes
`scroll`	When you scroll the page
`beforeunload`	When you're about to leave the page
`keydown`	When any key is pressed

--->NOTE-->without eventlistener we use on before an event like onmouseover but when we use event listner we do not

use on befor event only type mouseover

What is addEventListener?

It's a way to tell JavaScript to do something when an event happens.

Like:when you click a button

when you press a key

when the page loads

Simple Format:

```
element.addEventListener("eventName", functionToRun);
```

Why use addEventListener?

You can add multiple events to same element

Keeps JavaScript and HTML clean and separate

More flexible than using onclick or onkeydown directly

this in addEventListener :

In addEventListener, this means the element that triggered the event.

Example:

```
button.addEventListener("click", function() {  
    this.style.color = "red";  
});
```

Scroll Event

The scroll event happens when you scroll up or down on a page or inside a scrollable element.

Use Cases:

Show "Back to Top" button

Make sticky headers

Load more data when user scrolls to bottom (like Instagram)